Fiscal Year:	FY 2020	**Task Last Updated:**	FY 08/03/2020
PI Name:	Johnson, Matthew Ph.D.		
Project Title:	Intelligent Assistance for Accurate and Reliable Lymphatic Data Collection		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	mjohnson@ihmc.us	**Fax:**	FY
PI Organization Type:	NON-PROFIT	**Phone:**	
Organization Name:	Institute for Human and Machine Cognition		
PI Address 1:	40 South Alcaniz		
PI Address 2:			
PI Web Page:			
City:	Pensacola	**State:**	FL
Zip Code:	32502	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	TRISH--Synergy
Start Date:	03/01/2019	**End Date:**	03/31/2020
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 3/31/2020 per TRISH July report submission (Ed., 9/26/20)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Kernagis, Dawn Ph.D. (Florida Institute For Human And Machine Cognition Inc.) Crandall, David Ph.D. (Indiana University)		
Grant/Contract No.:	NNX16AO69A-SYN0006		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Synergy Project This project aims to use machine learning techniques to provide the user feedback about the usefulness and quality of the data being collected in terms of accurately measuring lymphatic function (Principal Investigator: Dawn Kernagis) in order to improve lymphatic function measurement.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	The overall impact anticipated by this work is improved technology designed to enhance crew independence, which will in turn improve space exploration. This technology also has the potential to provide better, cheaper health care on Earth by making complex tools easy to use and more accessible to non-experts.
Task Progress:	This project is aimed at augmenting the current Translational Research Institute for Space Health (TRISH) grant entitled Cervical Lymphatic Function Quantification and Associated Molecular Changes in Response to Simulated Microgravity. The relevant TRISH team members currently include Dr. Lihong Wang and his research team who are developing a lymphatic monitoring technology in the laboratory of California Institute of Technology (CalTech), Dr. Dawn Kernagis of the Florida Institute for Human and Machine Cognition (IHMC) who is conducting laboratory evaluation of the device, and Dr. Crandall of Indiana University (IU) who is quantifying lymphatic function using automatic techniques. This research will directly augment Dr. Crandall's effort captured as aim 3 of the modified research plan which is to develop visual quantification software for quantifying lymphatic function based on images captured by the device. Aim 3 leverages advances in machine learning techniques to improve the measurement of lymphatic function. While machine learning techniques have made great advances recently, they can suffer from brittleness due to their data dependency. This is compounded by the fact the data collection is performed by a human user who is generally unaware of the machine learning data quality requirements and may have other environmental conditions inhibiting the collection of clean usable data that would provide the best outcome for automated lymphatic function assessment. We propose to augment automated measurement research with a research effort targeting assistance of the user collecting the data. Our specific aim is to leverage the same automated techniques under development, but instead of automatic measurement, we will focus on providing intelligent assistance to the user collecting the data. Specifically, we aim to provide the user feedback about the usefulness of the data being collected in terms of accurately measuring lymphatic function. If successful, this will ensure that the data collected is useful for generating accurate automatic assessments. Our second goal is to provide feedback about sensor positioning to improve the usefulness of the data being collected. If successful, this would not only let the user know when the data is valid, but guide the user in positioning of the sensor to improve data quality. Results: - U-Net model seems appropriate and viable for bio-medical image analysis. - Lymphatic nodes may be insufficient for patient assessment. - Additional anatomical features can provide increased medical awareness and may improve automated recognition.
Bibliography Type:	Description: (Last Updated:)